Semiconductor Innovation Trends in European Patent Filings, Nov 2024 – Sep 2026

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Executive summary

Between November 2024 and mid-September 2026, **8,062 non-PCT semiconductor applications** (7,560 families) were first published at the EPO and at national offices of EPO member states. Chip integration and packaging, not transistors, is the largest field: it appears in 51% of all applications.

Most prominent technologies (absolute counts)

- **Power module and discrete packaging: 1,127 applications.** Led by Infineon (160), Nexperia (69), Intel (63), Bosch (62) and Mitsubishi Electric (52). This is Europe's automotive power-electronics base showing up in patents.
- **2.5D, chiplets, interposers and RDL: 896.** Intel dominates (158), ahead of Samsung (61), Infineon (60), TSMC (54) and NXP (47).
- **Gate-all-around / nanosheet FETs: 683.** Almost entirely Intel (233), Samsung (176) and TSMC (145), with imec (39) and CEA (23) as the European research players.
- **3D chip stacking and TSV: 620.** Intel (153) and Samsung (81) lead; up 33% between comparable periods.
- **Quantum processor architecture and qubit control: 502**, plus 478 on quantum algorithms. Fujitsu, Infineon, Alice & Bob, eleQtron and Pasqal each have around 30 in the hardware group.
- **Power transistors are Europe's home ground:** power MOSFETs 423 (Infineon 118), IGBTs 245 (Infineon 75, Mitsubishi Electric 65), HEMTs 208 (STMicroelectronics 36, Samsung 29, Infineon 22).

Most prominent players. Samsung (720 applications), Intel (560), Infineon (508), TSMC (352), STMicroelectronics (298) and CEA (288) together appear on 2,726 applications, 34% of the corpus. Intel and Samsung lead in advanced logic and packaging. Infineon, STMicroelectronics, Bosch, Mitsubishi Electric and Nexperia lead in power devices. CEA is the largest research filer, with a profile spanning ReRAM, spin qubits and process technology.

Momentum. Total filings were flat (−5% between Nov 2024–Aug 2025 and Nov 2025–Aug 2026). The fastest-growing technologies are CFET stacked transistors (10 → 29), SRAM (24 → 46), glass and ceramic package substrates (75 → 117, driven by Intel and Absolics),

quantum error correction (72 → 105) and 3D stacking (238 → 317). Front-end process categories such as lithography (−40%), doping (−36%) and deposition (−28%) declined.

Counts are full counts: one application can belong to several technologies, so categories do not add up to the total.

Main caveat. Excluding PCT-route filings removes several large players almost entirely: Qualcomm, IBM, Applied Materials, Huawei and Quantinuum file their European semiconductor applications mainly via PCT. The rankings show who files directly in Europe, not total European patent activity.

Scope and corpus

The corpus holds every application from the EPO or one of the 39 EPO member-state offices whose first publication fell between 1 Nov 2024 and the data cut-off (EP: 16 Sep 2026), that did not go through the PCT route, and that carries a CPC code in scope.

Measure	Value
Applications in scope	8,062
Patent families (DOCDB)	7,560
Direct EP applications	4,382 (54%)
National applications	3,680 (DE 2,590 · FR 464 · GB 259 · NL 127 · other 240)
Filed by companies only	7,138 (89%)
Filed by universities or research institutes only	672 (8%)
Joint company–research filings	114 (1%)
Excluded as out of scope (LED, solar cell or OLED packaging)	501

By technology domain (an application can sit in several):

Domain	Applications	Share	Nov 24–Aug 25	Nov 25–Aug 26	Change
Chip integration and packaging	4,081	51%	1,898	1,778	−6%
Transistors and devices	2,906	36%	1,290	1,316	+2%
Fabrication processes and equipment	1,857	23%	920	741	−19%
Memory	924	11%	409	434	+6%

Domain	Applications	Share	Nov 24–Aug 25	Nov 25–Aug 26	Change
Quantum	924	11%	404	403	0%
Chip design and layout	295	4%	135	138	+2%
All applications	**8,062**		**3,695**	**3,525**	**−5%**

Where the applicants come from (country of the first-named applicant): Germany 1,470 · USA 1,359 · Korea 1,084 · Japan 807 · Taiwan 569 · France 562 · Netherlands 391 · Switzerland 337 · China 273 · Austria 247. Applicants from EPO member states account for 3,134 applications (39%); 241 records carry no applicant country.

The two comparison windows are ten months each, Nov–Aug, because the data ends in mid-September 2026. A full-year comparison would understate year two by about six weeks.

Technology ranking

Thermal management, encapsulation and power-module packaging are the three largest categories, each with more than 1,100 applications. The largest transistor architecture is gate-all-around (683). Shares refer to the 8,062 applications in scope. "Change" compares Nov 2024–Aug 2025 with Nov 2025–Aug 2026; treat changes on fewer than about 30 applications as indicative only.

#	Technology	Domain	CPC basis	Applications	Families	Share	Change (P1 → P2)	Top 5 applicants (applications)
1	Thermal management	Integration	H10W40, H10W90/288	1,180	1,097	14.6%	557 → 507 (−9%)	Bosch 91 · Infineon 89 · ZF 71 · Intel 44 · Mitsubishi Electric 28
2	Encapsulation and device protection	Integration	H10W74, H10W42	1,154	1,080	14.3%	549 → 489 (−11%)	Infineon 160 · Intel 102 · STMicroelectronics 79 · NXP 78 · TSMC 62
3	Power module and discrete packaging	Integration	H10D80/2, H10W90/73–81, H10W76 + text	1,127	1,059	14.0%	499 → 508 (+2%)	Infineon 160 · Nexperia 69 · Intel 63 · Bosch 62 · Mitsubishi Electric 52
4	BEOL / on-chip interconnects	Integration	H10W20	1,093	1,039	13.6%	512 → 470 (−8%)	Samsung 224 · Intel 206 · TSMC 126 · STMicroelectronics 43 · Infineon 38

#	Technology	Domain	CPC basis	Applications	Families	Share	Change (P1 → P2)	Top 5 applicants (applications)
5	2.5D, chiplets, interposers and RDL	Integration	H10W70/61, H10W90/10, /40, /724 + text	896	848	11.1%	395 → 399 (+1%)	Intel 158 · Samsung 61 · Infineon 60 · TSMC 54 · NXP 47
6	CMOS / logic device integration (general)	Transistors	H10D84/01, /83, /85	853	821	10.6%	353 → 410 (+16%)	Intel 233 · Samsung 188 · TSMC 161 · imec 39 · Infineon 28
7	Wafer handling and fab equipment	Processes	H10P72	740	680	9.2%	351 → 299 (−15%)	Screen Holdings 42 · ASML 39 · Kokusai Electric 37 · TSMC 26 · LPE 21
8	Gate-all-around / nanosheet / nanowire FET	Transistors	H10D30/43, /50x, /014, /019x, /6735, H10D84/851–852 + text	683	668	8.5%	289 → 340 (+18%)	Intel 233 · Samsung 176 · TSMC 145 · imec 39 · CEA 23
9	3D chip stacking and TSV	Integration	H10W90/2x, H10W70/635 + text	620	587	7.7%	238 → 317 (+33%)	Intel 153 · Samsung 81 · TSMC 57 · Absolics 29 · NXP 26
10	Quantum processor architecture and qubit control	Quantum	G06N10/40 + text	502	460	6.2%	239 → 205 (−14%)	Fujitsu 32 · Infineon 32 · Alice & Bob 31 · eleQtron 30 · Pasqal 29
11	Quantum algorithms and software	Quantum	G06N10/20, /60, /80	478	453	5.9%	206 → 210 (+2%)	Fujitsu 74 · Multiverse Computing 22 · Alice & Bob 19 · Pasqal 18 · Tata Consultancy 15
12	Thin-film transistors (display, oxide TFT)	Transistors	H10D30/67xx, /031, H10D86/4, /6 + text	456	431	5.7%	200 → 217 (+9%)	LG 151 · Samsung 127 · BOE 41 · Intel 22 · TSMC 22
13	Deposition and layer formation (incl. epitaxy)	Processes	H10P14	438	401	5.4%	231 → 167 (−28%)	CEA 33 · TSMC 29 · Kokusai Electric 28 · Intel 25 · Soitec 21
14	Power MOSFET (trench, VDMOS, LDMOS, superjunction)	Transistors	H10D30/66, /65, /028–029, H10D84/141 + text	423	401	5.2%	192 → 173 (−10%)	Infineon 118 · Nexperia 35 · Bosch 34 · STMicroelectronics 32 · Mitsubishi Electric 29
15	Hybrid / direct bonding	Integration	H10W80, H10W90/79, H10P10 + text	413	390	5.1%	177 → 187 (+6%)	Samsung 65 · Intel 63 · TSMC 43 · Soitec 32 · CEA 29

#	Technology	Domain	CPC basis	Applications	Families	Share	Change (P1 → P2)	Top 5 applicants (applications)
16	Integrated passives (capacitors, resistors, inductors)	Transistors	H10D1, H10W20/49	396	378	4.9%	160 → 184 (+15%)	Samsung 68 · Intel 38 · CEA 24 · TSMC 24 · NXP 20
17	Silicon carbide (SiC)	Materials	H10D62/8325 + text	324	306	4.0%	154 → 134 (−13%)	Infineon 63 · Mitsubishi Electric 30 · STMicroelectronics 30 · Bosch 27 · Soitec 19
18	Wafer preparation and other processes	Processes	H10P90, H10P95	309	292	3.8%	159 → 117 (−26%)	Soitec 81 · CEA 46 · TSMC 13 · STMicroelectronics 11 · Infineon 9
19	FinFET	Transistors	H10D30/62, /024, H10D84/853 + text	303	300	3.8%	137 → 143 (+4%)	TSMC 108 · Samsung 80 · Intel 62 · Bosch 7 · imec 7
20	Etching	Processes	H10P50	292	268	3.6%	155 → 111 (−28%)	Shin-Etsu 39 · TSMC 28 · CEA 23 · SPTS 20 · Samsung 14
21	Gallium nitride / III-nitrides	Materials	H10D62/8503, /824 + text	290	254	3.6%	127 → 125 (−2%)	STMicroelectronics 44 · Infineon 29 · Bosch 28 · Samsung 28 · CEA 24
22	RF / impedance matching in packages	Integration	H10W44	284	265	3.5%	151 → 111 (−26%)	NXP 39 · Intel 30 · Infineon 21 · Samsung 12 · STMicroelectronics 11
23	Monolithic 3D integration	Integration	H10D88 + text	261	251	3.2%	111 → 134 (+21%)	Samsung 73 · Intel 48 · TSMC 47 · imec 29 · CEA 12
24	DRAM	Memory	H10B12 + text	251	248	3.1%	100 → 127 (+27%)	Samsung 166 · SK hynix 20 · TSMC 9 · imec 9 · Intel 8
25	Memory assemblies (HBM, memory–logic stacks)	Memory	H10B80 + text	251	243	3.1%	100 → 117 (+17%)	Samsung 115 · Intel 37 · TSMC 17 · SK hynix 15 · Kioxia 6
26	IGBT	Transistors	H10D12, H10D84/161 + text	245	240	3.0%	111 → 104 (−6%)	Infineon 75 · Mitsubishi Electric 65 · Nexperia 14 · Renesas 12 · STMicroelectronics 9

#	Technology	Domain	CPC basis	Applications	Families	Share	Change (P1 → P2)	Top 5 applicants (applications)
27	3D NAND / flash	Memory	H10B41, H10B43 + text	233	218	2.9%	108 → 99 (−8%)	Samsung 82 · YMTC 37 · SK hynix 21 · Kioxia 18 · STMicroelectronics 13
28	Oxide semiconductors (IGZO etc.)	Materials	H10D30/6755, H10D62/875 + text	225	212	2.8%	94 → 113 (+20%)	LG 75 · Samsung 57 · Semiconductor Energy Lab 13 · TSMC 11 · Intel 10
29	In-line testing and metrology	Processes	H10P74	217	199	2.7%	108 → 87 (−19%)	Samsung 20 · Infineon 17 · STMicroelectronics 12 · Bosch 10 · Intel 10
30	Glass and ceramic package substrates	Integration	H10W70/692 + text	209	203	2.6%	75 → 117 (+56%)	Intel 87 · Absolics 38 · Infineon 11 · Heraeus Electronics 6 · Samsung 6
31	HEMT (GaN / III-V)	Transistors	H10D30/471–475x, /015 + text	208	182	2.6%	91 → 93 (+2%)	STMicroelectronics 36 · Samsung 29 · Infineon 22 · CEA 17 · GlobalFoundries 14
32	Quantum error correction	Quantum	G06N10/70 + text	206	188	2.6%	72 → 105 (+46%)	Alice & Bob 26 · Fujitsu 24 · Riverlane 12 · Microsoft 10 · Xanadu 10
33	SiGe / Ge	Materials	H10D62/822 + text	204	203	2.5%	86 → 106 (+23%)	TSMC 69 · Samsung 52 · Intel 48 · GlobalFoundries 7 · IBM 3
34	CMP, grinding, cutting and dicing	Processes	H10P52–58	201	181	2.5%	95 → 87 (−8%)	STMicroelectronics 19 · Mitsubishi Electric 13 · TSMC 13 · Infineon 11 · CEA 10
35	Power and Schottky diodes	Transistors	H10D8 + text	191	175	2.4%	84 → 90 (+7%)	Infineon 33 · STMicroelectronics 26 · Mitsubishi Electric 22 · CEA 17 · Samsung 10
36	Standard-cell / device layout and routing	Design	H10D89/10, H10D84/9x	184	180	2.3%	71 → 97 (+37%)	Samsung 65 · TSMC 40 · Intel 27 · imec 10 · IBM 5
37	Doping (implantation, diffusion, irradiation)	Processes	H10P30–36	178	167	2.2%	94 → 60 (−36%)	Infineon 28 · STMicroelectronics 15 · Mitsubishi Electric 11 · TSMC 11 · Bosch 10

#	Technology	Domain	CPC basis	Applications	Families	Share	Change (P1 → P2)	Top 5 applicants (applications)
38	ESD and protection structures	Transistors	H10D89/6–9 + text	177	167	2.2%	92 → 67 (−27%)	Samsung 20 · STMicroelectronics 18 · NXP 17 · Infineon 16 · GlobalFoundries 13
39	Superconducting qubits / Josephson devices	Quantum	H10N60 + text	173	156	2.1%	75 → 77 (+3%)	Alice & Bob 26 · CEA 22 · Fujitsu 15 · CNRS 6 · FZ Jülich 5
40	ReRAM / phase-change memory	Memory	H10B63 + text	169	148	2.1%	89 → 72 (−19%)	CEA 36 · Samsung 31 · STMicroelectronics 22 · Micron 13 · TSMC 13
41	Backside power delivery	Integration	H10W20/481 + text	167	164	2.1%	88 → 65 (−26%)	Samsung 67 · Intel 59 · IBM 14 · TSMC 8 · imec 5
42	IC layout and physical design (EDA)	Design	G06F30/39x	150	148	1.9%	82 → 56 (−32%)	Samsung 23 · Intel 14 · TSMC 14 · Arteris 8 · imec 6
43	Lithography and masks	Processes	H10P76	129	122	1.6%	75 → 45 (−40%)	Shin-Etsu 50 · TSMC 13 · Intel 7 · imec 7 · CEA 6
44	FeRAM	Memory	H10B51, H10B53 + text	96	85	1.2%	50 → 38 (−24%)	Samsung 26 · CEA 23 · Intel 9 · imec 6 · TSMC 5
45	MRAM	Memory	H10B61 + text	89	86	1.1%	40 → 37 (−8%)	Samsung 21 · Everspin 10 · IBM 9 · CEA 7 · SK hynix 6
46	2D materials (graphene, TMDs)	Materials	H10D62/88x, H10D30/481, /017 + text	84	78	1.0%	40 → 40 (0%)	Samsung 16 · CEA 13 · Intel 13 · Paragraf 4 · TSMC 3
47	Semiconductor spin qubits / quantum dots	Quantum	H10D48/38x + text	83	61	1.0%	40 → 28 (−30%)	CEA 35 · TU Delft 12 · Quantum Technologies GmbH 5 · imec 3 · DLR 2
48	SRAM	Memory	H10B10 + text	75	75	0.9%	24 → 46 (+92%)	TSMC 27 · Intel 18 · Samsung 13 · imec 6 · IBM 4
49	SOI / engineered substrates	Transistors	H10D86/2, H10D87 + text	72	69	0.9%	29 → 38 (+31%)	GlobalFoundries 23 · STMicroelectronics 7 · Samsung 7 · CEA 6 · Intel 5
50	Bipolar transistors / HBT	Transistors	H10D10 + text	66	64	0.8%	34 → 28 (−18%)	GlobalFoundries 23 · STMicroelectronics 8 · Infineon 6 · NXP 6 · Ideal Power 3

#	Technology	Domain	CPC basis	Applications	Families	Share	Change (P1 → P2)	Top 5 applicants (applications)
51	Wafer cleaning	Processes	H10P70	63	59	0.8%	25 → 27 (+8%)	Screen Holdings 7 · TSMC 6 · Sumitomo Electric 5 · Infineon 4 · Kokusai Electric 3
52	Trapped-ion qubits	Quantum	text only	45	41	0.6%	28 → 11 (−61%)	Infineon 20 · eleQtron 16 · Univ. Siegen 5 · AQT 2 · Fraunhofer 2
53	CFET (stacked complementary FET)	Transistors	text only	43	42	0.5%	10 → 29 (+190%)	TSMC 15 · imec 11 · Intel 8 · CEA 5 · Huawei 3
54	Ferroelectric FET (FeFET)	Transistors	H10D30/701, /0415 + text	41	39	0.5%	16 → 20 (+25%)	Samsung 16 · Intel 6 · CEA 4 · imec 3 · GlobalFoundries 2
55	2D-material and carbon-nanotube FETs	Transistors	H10D30/481, /017 + text	39	37	0.5%	14 → 22 (+57%)	Samsung 14 · Intel 6 · CEA 5 · Huawei 2 · imec 2
56	Photonic quantum computing	Quantum	text only	25	24	0.3%	8 → 13 (+63%)	Quandela 3 · SaxonQ 3 · DLR 2 · PsiQuantum 2 · Q.ANT 2
57	Diamond	Materials	H10D62/8303	24	20	0.3%	10 → 11 (+10%)	CEA 5 · Mitsubishi Electric 5 · Diamfab 3 · Soitec 3 · Bosch 2
58	Fan-out and wafer/panel-level packaging	Integration	text only	15	15	0.2%	0 → 12	Silicon Box 5 · Infineon 2 · Samsung 2 · Absolics 1 · Hefei National Lab 1
59	Neutral-atom qubits	Quantum	text only	14	13	0.2%	7 → 6 (−14%)	Pasqal 5 · Max Planck Society 3 · LMU Munich 2 · others 1 each
60	Topological / Majorana qubits	Quantum	text only	8	8	0.1%	4 → 3	TU Delft 3 · Microsoft 2 · others 1 each
61	Gallium oxide (Ga_2O_3)	Materials	text only	6	6	0.1%	4 → 2	Bosch 3 · Disco 1 · Hangzhou Fujia Gallium 1 · Samsung 1

Company names are grouped at parent level (see Methodology). The application list with each application's categories and the basis of each assignment is in your BigQuery project (see Methodology).

Transistors and devices

Transistor filings split into two worlds with almost no overlap in applicants: advanced logic (GAA, FinFET, CFET) held by Intel, Samsung and TSMC, and power devices (power MOSFET, IGBT, HEMT, diodes) held by European and Japanese firms. The domain covers 2,906 applications and was stable (+2%).

Data

- **Advanced logic.** Gate-all-around is now larger than FinFET (683 vs 303 applications) and still growing (+18%); FinFET is flat (+4%). Intel, Samsung and TSMC hold 554 of the 683 GAA applications and 250 of the 303 FinFET applications. CFET, the next node after GAA, is still small (43) but almost tripled; TSMC (15) and imec (11) lead it.
- **Power devices.** Infineon leads power MOSFETs (118 of 423) and IGBTs (75 of 245), with Mitsubishi Electric second in IGBTs (65). In HEMTs (208), STMicroelectronics (36), Samsung (29), Infineon (22), CEA (17) and GlobalFoundries (14) are the leading filers. 186 of the 208 HEMT applications also carry the GaN/III-nitride material category, so HEMT here effectively means GaN HEMT. In power MOSFETs, 169 of 423 are SiC devices.
- **Display transistors.** TFTs (456) are a Korean field: LG (151) and Samsung (127) hold 61%, with BOE (41) third. Oxide-semiconductor (IGZO-type) channels grew 20%.
- **Beyond silicon.** 2D-material and carbon-nanotube FETs (39) and FeFETs (41) are small but growing; Samsung leads both.

Filing route (a pattern in the data): power-device filers use national offices heavily. Only 75 of 245 IGBT applications and 186 of 423 power-MOSFET applications are direct EP filings, against 411 of 683 for GAA. Bosch (2 of 247), Mitsubishi Electric (2 of 194) and Renesas (0 of 56) file almost exclusively at the German office, and so does TSMC (0 of 352).

Interpretation. Europe has no own player in leading-edge logic transistors; imec and CEA are the only European names among the top GAA filers, as research institutes. In power semiconductors Europe leads: Infineon, STMicroelectronics, Nexperia and Bosch are the main filers in every power category. The German-office route by TSMC and the Japanese power firms suggests that DE national filings, which become German patents faster and are cheaper, are used as the main European route in these fields.

Memory

Memory (924 applications, +6%) is Samsung's field: it files 66% of all DRAM applications (166 of 251) and 46% of memory-assembly applications (115 of 251). Europe's role is in emerging, embedded memories.

Data

- **DRAM and HBM-type assemblies** are the two largest memory categories (251 each) and both are growing (+27% and +17%). SK hynix is a distant second in DRAM (20).
- **3D NAND** (233) is the only large memory category in decline (−8%). Samsung (82) leads, followed by YMTC (37), SK hynix (21) and Kioxia (18).
- **SRAM** nearly doubled (24 → 46). TSMC (27) and Intel (18) lead it; these filings sit next to their GAA logic work.
- **Emerging memories** are smaller and declining: ReRAM/PCM 169 (−19%), FeRAM 96 (−24%), MRAM 89 (−8%). CEA leads ReRAM/PCM (36) and is second in FeRAM (23); STMicroelectronics is third in ReRAM/PCM (22).

Effect of the PCT exclusion. Memory makers that enter Europe via PCT are underrepresented here. Among EP semiconductor publications in the same window, Micron has 36 Euro-PCT applications against 21 direct ones, and CXMT, a Chinese DRAM maker, 52 against 2. Samsung files almost entirely direct (719 of its 720 applications here are direct EP), which inflates its weight in every Samsung-heavy category. *Interpretation:* Samsung's DRAM share would be lower in a count that includes Euro-PCT.

Chip integration, packaging and design

Integration is the largest domain (4,081 applications, 51% of the corpus) and has two separate halves: automotive power packaging led by European firms, and advanced logic packaging led by Intel, Samsung and TSMC. The domain declined 6% overall, but advanced packaging grew.

Data

- **Automotive and power packaging.** Thermal management (1,180), encapsulation (1,154) and power-module packaging (1,127) are the three largest categories in the whole study. Bosch (91), Infineon (89) and ZF (71) lead thermal management; ZF files 110 applications in total and Volkswagen 57, nearly all in packaging.
- **Advanced packaging.** 2.5D/chiplets/interposers (896) is Intel's strongest category (158). 3D stacking and TSV (620) grew fastest among the large categories (+33%). Hybrid bonding (413) is shared between Samsung (65), Intel (63), TSMC (43) and two French players, Soitec (32) and CEA (29).
- **Glass core substrates** are the fastest-growing packaging topic (75 → 117, +56%): Intel (87) and Absolics (38) account for most of it.
- **Backside power delivery** (167) is tied to GAA logic: 114 of these applications are also GAA filings. Samsung (67) and Intel (59) lead; the count fell 26%.
- **Monolithic 3D integration** grew 21% (261), with imec (29) as the leading non-Asian/non-US filer.

- **Chip design.** IC layout/EDA filings (G06F30/39, 150) fell 32%, while standard-cell and device-layout filings (184) rose 37%; Samsung, TSMC and Intel lead both. EDA tool vendors do not appear among the leaders; the only specialist in the top five is Arteris (8), an on-chip interconnect IP company.

Interpretation. Europe's integration strength lies in power and automotive packaging, where the main filers are European. In advanced logic packaging (chiplets, hybrid bonding, glass substrates), Europe's visible contributors are Soitec, CEA and imec, through wafer bonding and 3D-integration research. Fan-out packaging (15) is almost certainly undercounted: it has no own CPC group and is found by keyword only.

Quantum technologies

Quantum is the one domain where European start-ups and research institutes lead the filing ranks: after Fujitsu (98 applications), the top filers are CEA (57), Pasqal (36), Alice & Bob (34), Infineon (33) and eleQtron (31). The domain holds 924 applications and was flat between the two periods.

Data

- **Hardware vs software.** 630 applications concern quantum hardware (a qubit platform or G06N10/40 architecture and control); 253 are algorithms or error correction only. 319 could be assigned to a specific qubit platform.
- **Qubit platforms** (where identifiable):

Platform	Applications	Leading applicants
Superconducting (incl. cat qubits)	173	Alice & Bob 26 · CEA 22 · Fujitsu 15
Semiconductor spin / quantum dot	83	CEA 35 · TU Delft 12 · Quantum Technologies GmbH 5
Trapped ion	45	Infineon 20 · eleQtron 16 · Univ. Siegen 5
Photonic	25	Quandela 3 · SaxonQ 3 · DLR 2 · PsiQuantum 2 · Q.ANT 2
Neutral atom	14	Pasqal 5 · Max Planck Society 3 · LMU Munich 2
Topological / Majorana	8	TU Delft 3 · Microsoft 2

- **Error correction** is the fastest-growing quantum topic (72 → 105, +46%), led by Alice & Bob (26), Fujitsu (24) and Riverlane (12).

- **Other notable filers:** Multiverse Computing (23), TU Delft (20), Microsoft (18), DLR (17), Terra Quantum (17), Fraunhofer (15), Bull (14), IQM (14), Riverlane (14), Xanadu (14), Kipu Quantum (13).

Effect of the PCT exclusion. Several quantum companies file mainly via PCT and are underrepresented. Quantinuum, the largest trapped-ion company, has 50 Euro-PCT applications in the window and none filed directly, so it is absent here. IQM has 36 via PCT against 15 direct. *Interpretation:* the drop in trapped-ion filings (28 → 11) and the absence of Quantinuum say more about filing route than about the technology; Pasqal's lead in neutral atoms rests on very small numbers.

The platform counts depend on keywords in title and abstract, because CPC has no platform-specific groups except for superconducting (H10N60) and spin-qubit devices (H10D48/3835). German and French filings were searched with German and French terms.

Fabrication processes and equipment

Process and equipment filings (1,857) fell 19%, the steepest decline of any domain; eight of the nine process categories shrank. The field is led by equipment and materials suppliers rather than chipmakers.

Data

- **Equipment** (wafer handling and fab tools, 740) is led by Screen Holdings (42), ASML (39) and Kokusai Electric (37), with LPE (21), an Italian epitaxy-reactor maker, as the leading smaller European supplier.
- **Materials and wafers.** Shin-Etsu leads lithography and masks (50 of 129) and etching (39 of 292). Soitec leads wafer preparation (81 of 309) and is the fifth-largest filer in deposition (21).
- **Research.** CEA is the largest deposition filer (33) and appears in the top five of five process categories.

Effect of the PCT exclusion. Equipment and materials suppliers rely heavily on PCT. In the same window, Applied Materials has 131 Euro-PCT applications against 11 direct, Entegris 55 against 1, and NAURA 46 against 1. This domain is therefore the most incomplete in this study.

Interpretation. The decline may partly reflect classification timing: H10P only came into force with CPC 2026.01, and recently published documents may not yet carry their final H10P codes. A later re-run would show whether the drop holds.

Top 20 applicants

Samsung (720), Intel (560) and Infineon (508) are far ahead of everyone else; Intel is the fastest-growing large filer (+49%). Ten of the top 21 are European, including two research institutes (CEA, imec), two automotive suppliers (Bosch, ZF) and one carmaker (Volkswagen).

#	Applicant (group)	HQ	Applications	Families	Direct EP	Change (P1 → P2)	Main technologies (applications)
1	Samsung	KR	720	718	719	335 → 314 (−6%)	GAA 176 · DRAM 166 · TFT 127
2	Intel	US	560	552	332	196 → 293 (+49%)	GAA 233 · 2.5D/chiplets 158 · 3D stacking/TSV 153
3	Infineon	DE	508	488	135	227 → 216 (−5%)	Power-module packaging 160 · Power MOSFET 118 · IGBT 75
4	TSMC	TW	352	351	0	175 → 171 (−2%)	GAA 145 · FinFET 108 · SiGe 69
5	STMicroelectronics	CH/FR/IT	298	230	151	150 → 114 (−24%)	Power-module packaging 51 · GaN 44 · HEMT 36
6	CEA	FR	288	189	111	143 → 127 (−11%)	ReRAM/PCM 36 · Spin qubits 35 · Deposition 33
7	Bosch	DE	247	246	2	119 → 95 (−20%)	Power-module packaging 62 · Power MOSFET 34 · 2.5D/chiplets 31
8	Mitsubishi Electric	JP	194	194	2	76 → 97 (+28%)	IGBT 65 · Power-module packaging 52 · SiC 30
9	NXP	NL	169	169	169	90 → 70 (−22%)	Power-module packaging 50 · 2.5D/chiplets 47 · RF packaging 39
10	LG (Display, Electronics, Innotek)	KR	163	145	58	71 → 81 (+14%)	TFT 151 · Oxide semiconductors 75

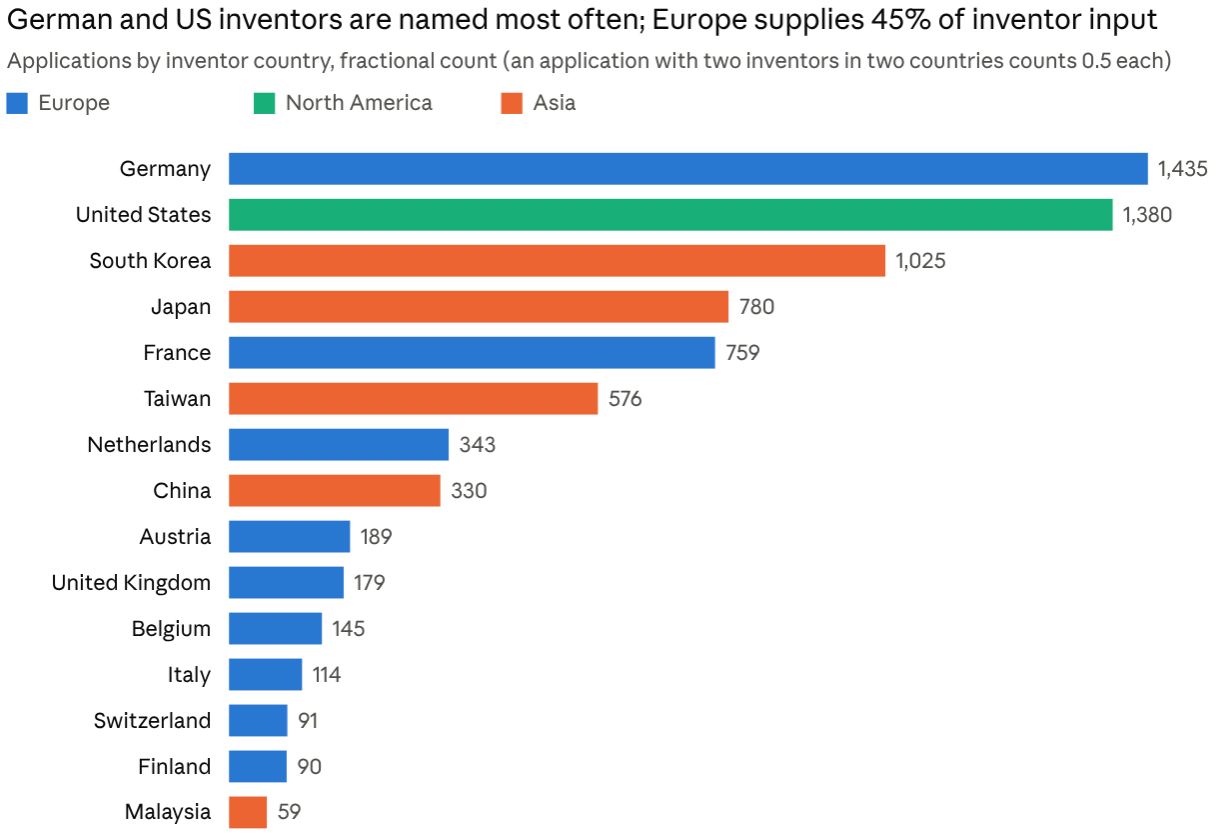
#	Applicant (group)	HQ	Applications	Families	Direct EP	Change (P1 → P2)	Main technologies (applications)
11	Nexperia	NL	154	130	113	64 → 83 (+30%)	Power-module packaging 69 · Power MOSFET 35 · 2.5D/chiplets 20
12	imec	BE	129	127	129	60 → 52 (−13%)	GAA 39 · Monolithic 3D 29 · Integrated passives 13
13	ZF Friedrichshafen	DE	110	109	7	74 → 28 (−62%)	Power-module packaging 45 · 2.5D/chiplets 23
14	Fujitsu	JP	101	101	101	20 → 60 (+200%)	Quantum algorithms 74 · Quantum processor/control 32 · Error correction 24
15	GlobalFoundries	US	96	95	93	42 → 42 (0%)	Bipolar/HBT 23 · SOI 23 · Integrated passives 17
16	Soitec	FR	93	91	3	40 → 33 (−18%)	Hybrid bonding 32 · Deposition 21 · SiC 19
17	IBM	US	67	67	4	53 → 14 (−74%)	GAA 18 · Backside power 14 · ReRAM/PCM 11
18	Shin-Etsu	JP	66	66	65	40 → 21 (−48%)	Lithography/masks 50 · Etching 39 · Deposition 9
19	SK hynix	KR	64	63	40	26 → 34 (+31%)	3D NAND 21 · DRAM 20 · HBM-type assemblies 15
20	Volkswagen	DE	57	44	16	15 → 34 (+127%)	Power-module packaging 39 · 2.5D/chiplets 16 · Glass substrates 5
20	Broadcom	US/SG	57	56	55	29 → 23 (−21%)	2.5D/chiplets 21 · 3D stacking/TSV 21 · Power-module packaging 11

"Main technologies" excludes the five catch-all categories (CMOS integration, BEOL, thermal, encapsulation, other wafer processes). STMicroelectronics and CEA have notably fewer families than applications because they file the same invention both

nationally (FR) and at the EPO. IBM's low direct-EP count reflects its use of the PCT route (138 Euro-PCT applications in the window).

Where the inventions are made

By inventor location, Europe contributes 45% of the corpus (3,536 applications, fractional count), Asia 37% (2,907) and North America 18% (1,433). Counted by applicant, Europe's share was only 39%: European labs produce more inventions than European companies own outright. Inventor countries are known for 7,912 of the 8,062 applications; Italian national filings carry no inventor data.



Google Patents Public Datasets (BigQuery), inventor_harmonized country codes, queried 2 Nov 2026 · 7,912 of 8,062 applications with inventor data · top 15 countries

Research is regional, not global. Only 370 applications (4.7%) name inventors from more than one world region; 3,399 are purely European, 2,784 purely Asian and 1,331 purely North American. Of the 3,658 applications with at least one European inventor, 3,353 (92%) have a European first-named applicant, 195 a North American and 71 an Asian one.

Share of inventor input by domain and technology (fractional, rows may not add to 100% because of rounding and other regions):

Domain / technology	Applications	Europe	Asia	N. America	of which DE	FR	AT
Quantum (domain)	894	71%	15%	13%	25%	17%	3%
Processes and equipment (domain)	1,805	50%	37%	12%	16%	13%	2%
Integration and packaging (domain)	4,030	43%	35%	22%	21%	8%	2%
Transistors and devices (domain)	2,867	31%	48%	21%	9%	8%	3%
Design (domain)	293	26%	43%	30%	9%	5%	1%
Memory (domain)	921	17%	63%	21%	1%	10%	0%
Trapped-ion qubits	45	99%	1%	0%	72%	0%	25%
Semiconductor spin qubits	82	90%	2%	7%	17%	43%	1%
Superconducting qubits	169	78%	12%	10%	15%	33%	1%
Gallium nitride	280	61%	23%	16%	15%	19%	5%
Silicon carbide	316	57%	36%	7%	25%	7%	11%
Power MOSFET	410	53%	37%	11%	26%	2%	13%
HEMT	201	52%	28%	21%	6%	17%	4%
Power-module packaging	1,106	49%	35%	16%	30%	5%	2%
IGBT	241	42%	53%	5%	21%	1%	12%
2.5D / chiplets	891	32%	34%	34%	16%	4%	3%
Hybrid bonding	408	32%	40%	28%	6%	14%	2%
3D stacking / TSV	619	18%	38%	44%	5%	4%	1%
GAA / nanosheet FET	683	13%	43%	44%	2%	4%	0%
FinFET	303	8%	58%	33%	2%	1%	1%
DRAM	251	5%	85%	10%	1%	1%	0%
Thin-film transistors	456	4%	88%	8%	1%	3%	0%

Inside the companies: where their inventors are (fractional share of each group's inventor input):

Applicant	Applications with inventor data	Inventor countries
Infineon	507	Germany 57% · Austria 25% · Malaysia 8% · Canada 3% · USA 2%

Applicant	Applications with inventor data	Inventor countries
STMicroelectronics	249	France 57% · Italy 31% · Singapore 3% (Italian filings lack inventor data)
NXP	169	Netherlands 91% · France 7%
Nexperia	154	UK 31% · Germany 22% · China 16% · Netherlands 11% · Hong Kong 11%
Bosch	247	Germany 93% · Hungary 4%
GlobalFoundries	96	USA 61% · Germany 14% · Singapore 13% · India 9%
Huawei	43	China 52% · Germany 43% · Belgium 5%
Texas Instruments	53	USA 62% · Germany 12% · Philippines 8%
onsemi	55	USA 41% · China 12% · Korea 11% · Czechia 7%
Intel	560	USA 93% · Germany 2%
Samsung	720	Korea 86% · USA 14%
TSMC	352	Taiwan 100%

Austria. 261 applications name at least one inventor in Austria. Infineon accounts for 188 of them (72%), followed by AT&S (23, all entities), TDK Electronics (10), Miba Sinter (8) and Alpine Quantum Technologies (4). Infineon's Austrian-invented applications centre on power semiconductors: power MOSFETs 81, SiC 50, IGBTs 44, power-module packaging 23, doping processes 21. They also include quantum work: qubit control 22 and trapped-ion 14. Of the 188, 95 were filed by Infineon Technologies Austria AG and 93 by the German parent, so counting by applicant entity alone would hide half of the Austrian output.

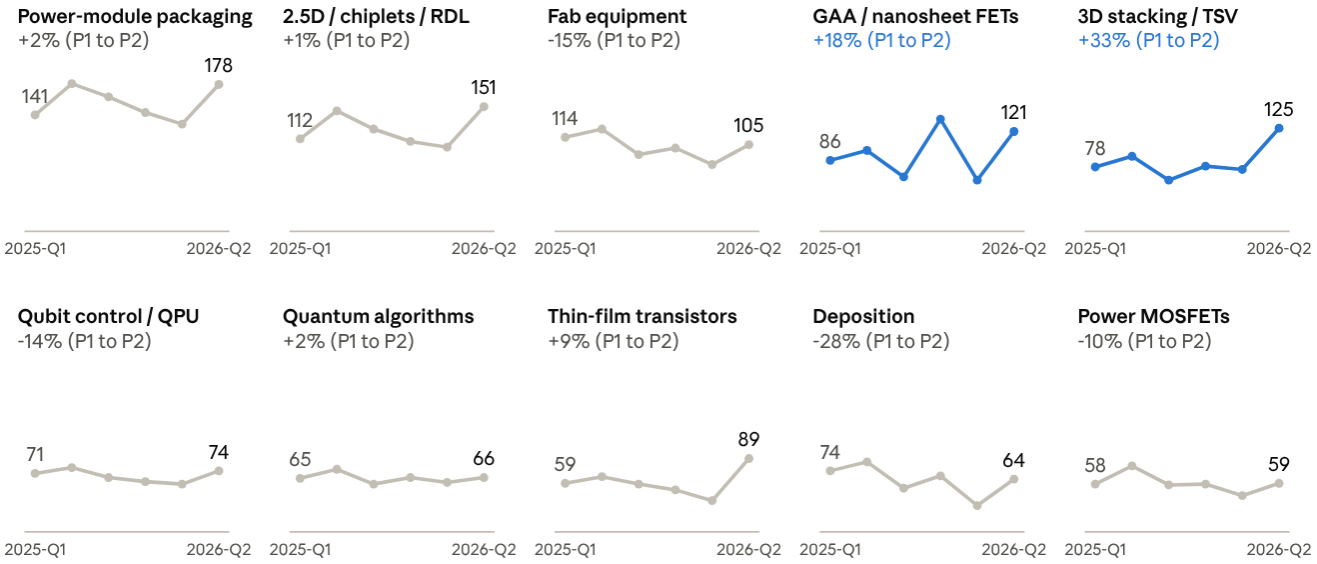
City level (e.g. Villach) is not in this dataset, which records only the inventor's country. The EPO Register has published each inventor's place of residence (city and postcode) since November 2021 (OJ EPO 2021, A12). For the EP applications, a Villach count is therefore possible as a second step. Note that it measures where inventors live, not where they work: Villach-site staff living in Klagenfurt would not show up as Villach.

Quarterly trends

Among the ten largest specific technologies, only gate-all-around transistors (+18%) and 3D stacking/TSV (+33%) rose clearly; both jumped in Q2 2026. Quarterly counts swing by up to a factor of two, so single quarters say little.

GAA transistors and 3D stacking are the only large technologies rising more than 10%

Applications per complete quarter, Q1 2025 to Q2 2026, same scale in every panel. Blue: up more than 10% between comparison periods.



Google Patents Public Datasets (BigQuery), queried 1 Nov 2026 · first publications, complete quarters Q1 2025 – Q2 2026 · change = Nov–Aug vs Nov–Aug

The partial quarters at either end are left out of the chart. Q4 2024 covers only November and December, and Q3 2026 only July to mid-September; together they hold 1,330 applications (553 and 777). The full quarterly series, including the partial quarters, is in the application list.

EP vs national filings

Direct EP applications make up 54% of the corpus (4,382); the German office alone adds another 2,590, which makes it the second channel for semiconductor filings in Europe. All other national offices together contribute 1,090.

Office	Applications	Nov 24–Aug 25	Nov 25–Aug 26	Data note
EP (direct, non-PCT)	4,382	1,937	1,959	Data to 16 Sep 2026
DE	2,590	1,172	1,149	
FR	464	207	203	

Office	Applications	Nov 24–Aug 25	Nov 25–Aug 26	Data note
GB	259	196	56	Drop most likely a coverage gap in the dataset
NL	127	46	75	
IT	77	54	15	No IT data after 21 Dec 2025
FI	57	23	29	
SE, LU, AT, PL, TR, CH, ES, BE, HU, DK, RS, CZ, PT, SM	106			20 or fewer each

The 8,062 applications form 7,560 families, so only 502 are further filings of an invention already counted. These cluster at STMicroelectronics (298 applications in 230 families; FR 98, IT 49, EP 151) and CEA (288 in 189; FR 177, EP 111), which file nationally first and then at the EPO.

The applicant mix differs sharply by route. Samsung, NXP, imec, Fujitsu and Shin-Etsu file almost only at the EPO. TSMC (all 352), Bosch (244 of 247), Mitsubishi Electric (191 of 194) and Renesas (all 56) file almost only at the German office, and Soitec at the French office (90 of 93). Infineon (370 DE, 135 EP) and Intel (219 DE, 332 EP) use both routes.

Emerging technologies

Seven small technologies grew by 30% or more. Most point to the logic generation after GAA (CFET, SRAM scaling, cell layout) and to packaging for AI chips (glass substrates). Criteria: at least 10 applications in the first period, at least +30% growth, under 300 applications in total.

Technology	Nov 24–Aug 25 → Nov 25–Aug 26	Change	Leading applicants
CFET (stacked complementary FET)	10 → 29	+190%	TSMC · imec · Intel
SRAM	24 → 46	+92%	TSMC · Intel · Samsung
2D-material and carbon-nanotube FETs	14 → 22	+57%	Samsung · Intel · CEA

Technology	Nov 24–Aug 25 → Nov 25–Aug 26	Change	Leading applicants
Glass and ceramic package substrates	75 → 117	+56%	Intel · Absolics
Quantum error correction	72 → 105	+46%	Alice & Bob · Fujitsu · Riverlane
Standard-cell / device layout	71 → 97	+37%	Samsung · TSMC · Intel
SOI / engineered substrates	29 → 38	+31%	GlobalFoundries · STMicroelectronics

Below the 10-application threshold, photonic quantum computing (8 → 13) and fan-out packaging (0 → 12) also rose. Among large technologies, 3D stacking/TSV (+33%) is the only one growing at a similar rate.

Interpretation: the group of CFET, SRAM and cell-layout filings by TSMC, Intel and Samsung looks like early preparation for nodes beyond the first GAA generation. Error-correction filings rising while qubit-hardware filings fall (−14%) fits a field moving toward fault-tolerant architectures. Backside power delivery, often named as a next-node technology, fell 26%; its main filing wave may already lie before this window.

Methodology and limitations

All counts come from the Google Patents Public Datasets table `patents-public-data.patents.publications`, queried on 1 Nov 2026 from BigQuery project `patent-study-513418`. Latest publication dates in the data: EP 16 Sep 2026, DE 24 Sep 2026, GB 16 Sep 2026, FR 4 Sep 2026.

Selection

1. Offices: EP plus the 39 EPO member states.
2. An application is counted once, at its first publication; it is in the window if that date is 1 Nov 2024 – 31 Oct 2026.
3. Excluded: anything with a PCT number (Euro-PCT and PCT national phase), national translations of EP patents (kind T1–T5), utility models (U), and A4 supplementary search reports.
4. Scope by CPC (any position, not only the first code): H10D, H10B, H10W, H10P, H10N60, G06N10, G06F30/39, B82Y10 (B82Y10 only together with a quantum keyword). Legacy H01L 21/23/25/27/29 was also searched, but no document in the window still carries it:

CPC 2025.01 and 2026.01 moved devices to H10D and packaging and processes to the new H10W and H10P.

5. Out of scope: 501 applications that entered only through H10W/H10P and concern LEDs, solar cells or OLEDs (H10H, H10F, H10K, without any device or quantum code).
6. Duplicates: where an application had several publications on its first date, one record was kept. Families are DOCDB family IDs.

Classification

- 61 categories, each defined by CPC regular expressions and, where useful, keywords in title and abstract (English, German, French). Keywords are only applied to applications already in the matching CPC domain. All rules are stored in the table `patent_study.rules`.
- Every assignment records its basis: CPC, CPC+text, or text only. Of 18,371 assignments, 16,270 rest on CPC alone, 1,402 on CPC confirmed by text and 699 on text only (96% CPC-based); seven categories are text-only (CFET, fan-out, trapped ion, neutral atom, photonic, topological, Ga2O3).
- Full counting: an application is counted in every category it matches. 334 applications fall into a domain but no specific category (e.g. only electrode or connector codes).
- Spot check: a sample of text-based and CPC-based assignments was read. One error was found and fixed: H10D30/47 also covers nanoribbon FETs, so HEMT was restricted to H10D30/471 ff. and H10D30/015.

Applicants

- Starting point is Google's harmonised assignee name. Name variants and subsidiaries were merged into 51 groups, e.g. all Infineon entities, Samsung Electronics + Samsung Display + Samsung Electro-Mechanics, LG Display + LG Electronics + LG Innotek, BOE and its subsidiaries, Kia + Hyundai Motor + Hyundai Mobis, the two CEA name variants. Co-applicants each receive full credit.
- HQ countries in the top-20 table are the group's headquarters; the origin figures in Scope use the first-named applicant's country.
- Applicant type is rule-based (company / university or research / other); 138 applications by unsuffixed names could not be classified.

Known limitations

- **PCT exclusion biases the picture.** In the same window, Qualcomm (188), Huawei (140), IBM (138), Applied Materials (131), BOE (83), Google (74), Microsoft (71), CXMT (52) and Quantinuum (50) filed semiconductor applications at the EPO mainly or only via PCT. Their weight here is far below their real European activity.
- **National coverage is uneven.** Italian data ends in Dec 2025, British data thins out sharply in 2026, and small offices are sparsely covered.

- **Text-only categories** depend on keywords and probably undercount, especially for national filings in other languages.
- **Publication, not filing.** The window is by first publication, which lags filing by about 18 months; the trends describe inventions filed roughly in 2023–2025.

Data and cost

- BigQuery processed about 610 GB in total (two corpus builds of 263–264 GB each; the first build missed H10W/H10P and was redone), within the free 1 TB/month.
- Tables in `patent-study-513418.patent_study`: `corpus` (raw selection), `apps_final` (8,062 applications in scope), `rules`, `cats` (application × category with basis), `appl` (applicant grouping), `application_list` (the full list, one row per application).
- **Export before about 31 Dec 2026**: tables in a BigQuery sandbox expire 60 days after creation. In the BigQuery console, query `SELECT * FROM patent_study.application_list` and use Save results → CSV (Google Drive).

The final selection query:

```
CREATE OR REPLACE TABLE `patent-study-513418.patent_study.corpus` AS
WITH first_pub AS (
  SELECT application_number, MIN(publication_date) AS first_publication_date
  FROM `patents-public-data.patents.publications`
  WHERE country_code IN
('EP','AL','AT','BE','BG','CH','CY','CZ','DE','DK','EE','ES','FI','FR','GB','
GR','HR','HU','IE','IS','IT','LI','LT','LU','LV','MC','ME','MK','MT','NL','NO
','PL','PT','RO','RS','SE','SI','SK','SM','TR')
  GROUP BY application_number
)
SELECT p.publication_number, p.country_code, p.kind_code,
p.application_number,
  p.family_id, p.publication_date, f.first_publication_date, p.filing_date,
p.priority_date,
  (SELECT t.text FROM UNNEST(p.title_localized) t ORDER BY t.language = 'en'
DESC LIMIT 1) AS title,
  (SELECT a.text FROM UNNEST(p.abstract_localized) a ORDER BY a.language =
'en' DESC LIMIT 1) AS abstract,
  ARRAY(SELECT c.code FROM UNNEST(p.cpc) c) AS cpc_all,
  ARRAY(SELECT c.code FROM UNNEST(p.cpc) c WHERE c.first) AS cpc_first,
  ARRAY(SELECT c.code FROM UNNEST(p.cpc) c WHERE c.inventive) AS
cpc_inventive,
  ARRAY(SELECT a.name FROM UNNEST(p.assignee_harmonized) a) AS applicants,
  ARRAY(SELECT a.country_code FROM UNNEST(p.assignee_harmonized) a) AS
applicant_countries
FROM `patents-public-data.patents.publications` p
JOIN first_pub f USING (application_number)
```

```
WHERE f.first_publication_date BETWEEN 20241101 AND 20261031
  AND p.publication_date = f.first_publication_date
  AND (p.pct_number IS NULL OR p.pct_number = '')
  AND p.kind_code NOT IN ('T','T1','T2','T3','T4','T5','U','U1','U8','A4')
  AND EXISTS (SELECT 1 FROM UNNEST(p.cpc) c WHERE REGEXP_CONTAINS(c.code,

r'^(H10D|H10B|H10W|H10P|H10N60/|H01L(21|23|25|27|29)/|G06N10/|B82Y10/|G06F30/
39)'))
```